Astana IT University · School of Software Engineering

Managing the development of an MVP web platform for Unified National Testing (UNT) preparation in Computer Science and assessing its commercialization potential

Diploma project (individual, startup track): consolidated report for probationary period 1

Student	Yerzat Qaldyqan, group SE-2408 (GitHub: Yerrrzat)
Supervisor	Zarina Almabekkyzy
Repository	https://github.com/Yerrrzat/Web-platform-UNT- (tag `prob-period-1`)
Prototype	https://yerrrzat.github.io/Web-platform-UNT-/prototype/ (source: prototype/index.html)
Diagram sources	diagrams/src/*.puml in the repository
YouTrack	https://yerzat-diploma.youtrack.cloud/projects/UNT (project key UNT)
Date	October 2026, Astana

Table of contents

1. Project Proposal
- 1.1 Problem statement
- 1.2 Target users
- 1.3 Aim
- 1.4 Objectives
- 1.5 Expected outcomes
- 1.6 Scope
- 1.7 Engineering complexity
- 1.8 Assumptions and constraints

2. Evidence of Relevance
- 2.1 Evidence sources
- 2.2 Findings from data and observation (E1–E3)
- 2.3 Interview guide (used for E4)
- 2.4 Interview results (E4)
- 2.5 Verified findings vs. assumptions
- 2.6 Impact on requirements

3. Literature Review
- 3.1 Method
- 3.2 Theme A: Practice testing improves learning
- 3.3 Theme B: Feedback design
- 3.4 Theme C: Engagement and motivation
- 3.5 Theme D: UNT and digital preparation in Kazakhstan
- 3.6 Theme E: Freemium monetization
- 3.7 Official and industry sources
- 3.8 Structured analysis of publications
- 3.9 What the review means for the project
- References

4. Existing System Comparison
- 4.1 Selection of systems
- 4.2 Comparison criteria
- 4.3 Comparison table
- 4.4 Analysis
- 4.5 Why the proposed solution is needed

5. Technology Comparison
- 5.1 Decision 1: Web application framework (frontend + server)
- 5.2 Decision 2: Data storage
- 5.3 Decision 3: Component interaction and API style
- 5.4 Decision 4: Payments (subscription)
- 5.5 Supporting choices (not weighted)

6. Requirements Specification (initial)
- 6.1 Actors
- 6.2 Functional requirements
- 6.3 Non-functional requirements
- 6.4 Constraints
- 6.5 Traceability summary

7. Architecture and Diagrams
- 7.1 Architectural overview
- 7.2 System context diagram
- 7.3 Use case diagram

7.4 Component diagram (draft)
7.5 Data model (draft)
7.6 Key scenario: taking a test (sequence, textual)
7.7 Alignment check

8. UX / Prototype
8.1 Prototype links
8.2 Key user scenarios
8.3 Screens
8.4 Example input data: version import format (JSON)
8.5 Planned usability checks

9. Verification and Evaluation Plan
9.1 Verification (testing plan)
9.2 Evaluation: outcome metrics
9.3 Measurement conditions
9.4 Baseline for comparison
9.5 Commercialization assessment
9.6 Threats to validity

10. Project Management and Risks
10.1 Tools and links
10.2 Roles and responsibilities
10.3 Way of working
10.4 Milestones
10.5 Backlog snapshot (top items)
10.6 Risk register
10.7 Dependencies
10.8 Next steps (after prob-period-1)

11. Supervisor Feedback
11.1 Feedback log
11.2 Agreed checkpoints

1. Project Proposal

Item	Details
Project title	Managing the development of an MVP web platform for Unified National Testing (UNT) preparation in Computer Science and assessing its commercialization potential
Working product name	UNT Informatics Prep (repository: `Web-platform-UNT-`)
Type	Individual diploma project (startup track)
Student	Yerzat Qaldyqan, group SE-2408, Astana IT University
Supervisor	Zarina Almabekkyzy
Stage	Probationary period 1 (prob-period-1), October 2026

1.1 Problem statement

Every year, school leavers in Kazakhstan sit the Unified National Testing (UNT, ҰБТ / ЕНТ). It is a computer-based exam of 120 items over five subjects, lasts four hours and is scored out of 140 points. Students on the *Mathematics – Informatics* profile answer 40 Informatics items, so this subject alone carries a large share of the score that decides state grants for IT programmes.

Several commercial preparation services exist (iTest, Joo.kz, Daryn.online and others; see Section 4), and the National Testing Center (NTC) runs a free trial test. However, the following problems remain for Informatics specifically:

1. **Informatics is a minor subject on general platforms.** Large services focus on the compulsory subjects and on mathematics. Informatics content is thin, rarely updated to the current specification, and often lacks Python code and SQL table tasks, which are now standard UNT item types.
2. **Weak feedback after a mock test.** Many services show only a score or the correct letter. Students cannot see *why* an answer is wrong or which specification topic it belongs to, so they do not know what to revise.
3. **Kazakh-language content gap.** Good Informatics material in Kazakh is scarcer than in Russian.
4. **Price and format mismatch.** Full courses cost tens of thousands of tenge and bundle live lessons that many students do not need. Students who only want realistic Informatics practice tests with explanations have no cheap, focused option.

1.2 Target users

User group	Description	Main need
Primary: UNT candidates (grades 10–11) on the Mathematics–Informatics profile	Kazakh- and Russian-medium school students preparing for the UNT	Realistic timed Informatics mock tests, explanations, topic-level weaknesses
Secondary: informatics teachers and tutors	School teachers and private tutors	A ready bank of specification-aligned tests to recommend to students

User group	Description	Main need
Business stakeholder: platform owner/administrator	The author as content owner and operator	Managing the test bank, subscriptions and usage analytics

1.3 Aim

To plan, manage and deliver a working MVP of a web platform where students take UNT-format Informatics mock tests with detailed explanations under a freemium subscription model, and to assess the platform’s commercialization potential using MVP usage and payment-intent data.

1.4 Objectives

#	Objective	Measurable result
O1	Confirm the problem with target users	3–5 structured interviews with students/teachers; a summary of verified findings vs. assumptions (Section 2)
O2	Analyse existing solutions and literature	≥ 12 sources (≥ 8 scientific); comparison of 5 systems against 8 criteria (Sections 3–4)
O3	Specify and design the MVP	Prioritised requirements with acceptance criteria; context, use case and component diagrams (Sections 6–7)
O4	Build the MVP	Deployed web application: registration, test catalogue, timed test runner, results with explanations, subscription paywall, admin test upload
O5	Fill the platform with content	≥ 10 complete 40-item Informatics versions prepared from the author’s existing UNT Informatics practice versions (2026 specification)
O6	Run a pilot and evaluate	≥ 30 pilot users; measure test completion rate, score improvement between attempts, SUS usability score and free-to-paid intent (Section 9)
O7	Assess commercialization potential	Unit economics (pricing, costs, break-even), market size estimate and go-to-market plan
O8	Manage the project traceably	YouTrack backlog, Git history linked to tasks, risk register, milestone reports (Section 10)

1.5 Expected outcomes

- A deployed MVP web application (Next.js + PostgreSQL) with source code and tests in this repository.
- An Informatics item bank in UNT format (single-choice, multiple-choice, matching and context-based items with Python code and SQL tables) in Kazakh, with Russian added later.
- A pilot evaluation report with usage metrics and user feedback.
- A commercialization assessment: business model canvas, pricing, unit economics and next steps.
- Project management artefacts: backlog, roadmap, risk register and change log.

1.6 Scope

In scope (MVP)

- Student registration and login (email + password).
- Catalogue of Informatics mock-test versions (40 items each, UNT structure).
- Timed test runner that imitates the UNT interface: navigation between items, flagging and auto-submit on timeout.

- Automatic scoring following UNT rules (1 point for single choice; partial credit for multiple-choice and matching items).
- Results page with the correct answer, an explanation per item, and a breakdown by specification topic.
- Attempt history and progress chart.
- Freemium model: a limited number of free versions, with full access through a paid subscription (payment provider integration in test mode for the MVP).
- Admin panel: import test versions (JSON/CSV with images), publish and unpublish, view basic statistics.
- Kazakh interface (Russian interface as a stretch goal).

Out of scope (for this diploma)

- Other UNT subjects (mathematics, history, etc.).
- Live lessons, video courses and tutoring.
- Native mobile apps (the web app is responsive instead).
- Adaptive or AI-generated questions (named as future work).
- Official affiliation with the National Testing Center.

1.7 Engineering complexity

The project goes beyond a simple CRUD website:

- **Domain-specific scoring engine:** UNT partial-credit rules for multiple-choice and matching items, and time-limited sessions that are resistant to page reloads.
- **Rich item rendering:** syntax-highlighted Python code, SQL tables and spreadsheet fragments, plus an import pipeline with validation for item banks.
- **Payments and access control:** subscription states, webhooks and entitlement checks on the server side.
- **Analytics:** per-topic mastery calculation and an event log for the commercialization metrics.
- **Non-functional quality:** performance under concurrent test sessions, security (OWASP Top 10) and automated testing in CI.

1.8 Assumptions and constraints

- Single developer and project manager, working part-time alongside coursework, from October 2026 to May 2027.
- Content is written by the author, who already writes UNT Informatics practice versions to the 2026 specification. Items are original and are not copied from NTC materials.
- Budget is limited to free tiers or low-cost hosting (≤ 10,000 KZT per month during the pilot).

2. Evidence of Relevance

This section separates **verified findings** (backed by data, documents or interviews) from **working assumptions** that still have to be tested during the pilot.

2.1 Evidence sources

#	Source type	What was done	Status
E1	Analysis of available data	Official UNT 2026 format: 120 items, 5 subjects, 4 hours, 140 points; Informatics is a profile subject paired with Mathematics with 40 items (Informburo, 2026; NTC specification)	Verified
E2	Analysis of available data	Market scan of UNT preparation services and their prices (Informburo, 2021; current websites of iTest, Joo.kz, Daryn.online, Baimly, NTC trial test), see Section 4	Verified
E3	Observation of the relevant process	Author's own experience of writing UNT Informatics practice versions to the 2026 specification and teaching Python to school students: recurring student errors and topics that need explanation	Verified (first-hand)
E4	Interviews with potential users	3–5 semi-structured interviews with UNT candidates (Mathematics–Informatics profile) and informatics teachers/tutors, using the guide in 2.3	**To complete before submission; results go in 2.4**
E5	Discussion with supervisor	Review of the topic, scope and the startup format with the supervisor (see Section 11)	**To record in Section 11**

2.2 Findings from data and observation (E1–E3)

1. **Informatics carries a large share of the score for IT applicants.** In the 2026 format, each profile subject has 40 items. For students on the Mathematics–Informatics profile, Informatics is one of the two subjects that most affect the grant score. *(E1)*
2. **A free official trial exists but only gives a score and a topic analysis.** The NTC trial (prob-ent.testcenter.kz) is free and gives an individual thematic analysis. No source says it explains individual answers. *(E1, E2)*
3. **Commercial services are bundled and expensive, or general-purpose.** Course-based services cost from about 9,900 to over 100,000 KZT per course (2021 prices). Test-bank services such as iTest are cheap (from 4,490 KZT per 90 days) but cover 20 subjects with general content. None of the reviewed services is focused on Informatics. *(E2)*
4. **Item types have changed.** Current UNT Informatics versions use context blocks with Python code, SQL queries over tables and spreadsheet fragments. Producing such items with good distractors takes expertise and time, which is why generic banks lag behind. *(E3)*
5. **Students need explanations, not just answers.** When working with students, the author sees the same error patterns repeatedly: mixing up units of information, off-by-one errors in loops, and SQL condition logic. A correct letter alone does not fix these errors; a short explanation tied to the topic does. *(E3; to be confirmed in E4)*

2.3 Interview guide (used for E4)

Participants: 3–5 people in total: at least 2 grade 10–11 students on the Mathematics–Informatics profile, and at least 1 informatics teacher or tutor. **Format:** 15–20 minutes, in person or by call, in Kazakh or Russian. Notes are anonymised (P1…P5); no personal data is stored in the repository.

Students 1. How are you preparing for the Informatics part of the UNT now? Which resources do you use? 2. How often do you take full mock tests? Where? 3. After a mock test, what do you do with your mistakes? What information is missing? 4. Which Informatics topics are hardest for you (Python code,

SQL, spreadsheets, number systems, networks…)? 5. Have you paid for any UNT preparation? How much? What made it worth paying for, or not? 6. What would make you choose a new Informatics-only platform? What price per month seems fair? 7. Do you prefer the phone or a computer for practising tests?

Teachers / tutors 1. How do you prepare students for UNT Informatics? Which materials do you use? 2. What problems do you see with existing test banks (outdated, errors, no explanations, language)? 3. Would you recommend a paid platform to students? Under what conditions? 4. What would you need to see (reports, assigning tests) to use it with a class?

2.4 Interview results (E4)

> **Status (October 2026):** interviews are being scheduled with students of the Mathematics–Informatics profile and informatics teachers. Results will be added as anonymised notes after each interview (task UNT-3).

ID	Role / grade	Language	Date	Key statements (short)	Supports finding #	Contradicts / new insight
P1						
P2						
P3						
P4						
P5						

Summary of interview findings: will be added after the interviews are completed (UNT-3).

2.5 Verified findings vs. assumptions

Statement	Status	Evidence / how it will be tested
UNT 2026 has 40 Informatics items for the Mathematics–Informatics profile	**Verified**	E1
Free official trial tests exist (NTC)	**Verified**	E1, E2
No reviewed service focuses on Informatics with item-level explanations	**Verified for the 5 reviewed systems**	E2, Section 4
Students value item-level explanations more than extra test volume	Assumption	E4 interviews; pilot survey (Section 9)
Students are willing to pay about 1,500–2,500 KZT per month for an Informatics-only service	Assumption	E4 question 5–6; pilot paywall click-through and pre-orders

Statement	Status	Evidence / how it will be tested
Teachers would recommend the platform to classes	Assumption	E4 teacher interviews; pilot referral count
Kazakh-language content is a differentiator	Partly verified (author's observation)	E4; share of users choosing Kazakh in pilot

2.6 Impact on requirements

Finding	Requirement (Section 6)
UNT format and item types	FR-05 (timed runner), FR-06 (scoring rules), FR-07 (code/SQL rendering)
Need for explanations	FR-08 (explanations per item), FR-09 (topic breakdown)
Price sensitivity, freemium hypothesis	FR-11 (free tier), FR-12 (subscription), FR-15 (paywall analytics)
Kazakh-language gap	NFR-07 (Kazakh interface and content)

3. Literature Review

3.1 Method

Sources were found through Google Scholar, publisher sites and Kazakhstani university journals using these search terms: *testing effect / retrieval practice*, *practice tests feedback*, *gamification learning meta-analysis*, *Unified National Testing Kazakhstan*, *distance learning platform Kazakhstan*, *freemium willingness to pay*. Inclusion criteria:

- peer-reviewed journal articles or official documents;
- direct relevance to one of four themes: (A) how practice testing affects learning, (B) feedback design, (C) the UNT and digital preparation in Kazakhstan, (D) freemium business models;
- core sources published from 2020 onwards. Three foundational works (2006, 2008, 2013) are kept because later studies build on them.

The review covers **17 sources: 13 peer-reviewed scientific publications and 4 official or industry sources.**

3.2 Theme A: Practice testing improves learning

The central idea behind the platform is the **testing effect**: retrieving information from memory during a test strengthens long-term retention more than re-reading does. Roediger and Karpicke [1] showed this in controlled experiments. Dunlosky et al. [2] reviewed ten learning techniques and rated *practice testing* as one of only two with “high utility”.

Recent meta-analyses confirm that the effect holds in real classrooms, not only in laboratories. Adesope, Trevisan and Sundararajan [3] analysed 272 effect sizes and found that practice tests outperform restudy and other strategies, with stronger effects when the practice and final test formats match. Yang et al. [4] reviewed 222 classroom studies and found a consistent benefit of quizzing. Agarwal, Nunes and Blunt [5] reviewed applied studies in schools and also report consistent benefits across ages and subjects.

Implication for the project: mock tests should copy the real UNT format (item types, timing, interface) so that practice transfers to the exam (format match [3]). The platform should also encourage *repeated* attempts spread over time rather than one cramming session [2], [5].

Limitation taken into account: most studies measure retention of factual or conceptual knowledge. Benefits for procedural skills such as tracing Python code are less studied. The project therefore measures improvement directly in the pilot (Section 9) instead of assuming it.

3.3 Theme B: Feedback design

Testing without feedback leaves errors uncorrected. Shute [6] concludes that formative feedback works best when it is **elaborated** (explains *why*), specific and given soon after the response. Wisniewski, Zierer and Hattie [7] re-analysed 435 studies and found an average effect of $d \approx 0.48$. Feedback that contains high information (task-, process- and self-regulation-level) was far more effective than simple right/wrong signals.

Implication: every item needs an explanation and a topic tag, and the results page should show which *process* failed (for example, “loop boundary” rather than just “wrong”). This directly supports FR-08 and FR-09.

Limitation: elaborated feedback costs content effort. The MVP limits explanations to short texts written by the author and leaves AI-generated feedback for future work.

3.4 Theme C: Engagement and motivation

Sailer and Homner [8] found small but significant positive effects of gamification on cognitive, motivational and behavioural outcomes, with larger effects when game elements are combined with competition *and* collaboration. The effects were less stable in studies with strong designs.

Implication: the MVP uses only lightweight progress elements (streaks and a progress chart). It does not build complex gamification, because the evidence is moderate and the cost is high.

3.5 Theme D: UNT and digital preparation in Kazakhstan

Abdrasilov et al. [9] provide validity evidence for the UNT. The mathematics component shows strong psychometric properties, and UNT scores predict first-semester university performance. This confirms that the UNT is high-stakes and that preparation matters. Mingisheva [10] analyses UNT results from 2014, 2017 and 2020 and finds gaps between urban and rural schools and between Kazakh- and Russian-medium schools. This supports offering quality material in Kazakh and an online channel that reaches rural students.

Useinov et al. [11] describe preparing students in small rural schools for the UNT through distance learning combined with online test platforms, and report positive progress. Rakhmetov et al. [12] show that Kazakhstani informatics teachers are positive about distance-learning platforms (96% in their survey) but often lack the skills to build and use them. This points to a ready-to-use platform for teachers as a secondary market.

Limitations: [10] and [11] are regional studies with small samples, and none of the Kazakhstani sources evaluates Informatics preparation specifically. This gap is part of the project's motivation.

3.6 Theme E: Freemium monetization

Gu, Kannan and Ma [13] ran a randomized field experiment on a freemium publisher (free PDFs, paid books). The free version exerts a strong "zero-price" pull, but adding premium tiers increased sales of the existing premium product by 8.9–21.5% through compromise and attraction effects, although a cheap premium tier can cannibalize the main one. Tyrväinen and Karjaluoto [14] meta-analysed 55 studies and found that functional, hedonic, social and price value of the free tier predicts willingness to pay, partly through trust, and that the effect depends on the monetization strategy.

Implication: the platform offers two premium tiers (monthly and a cheaper-per-month "until UNT" pass) to use the effects in [13]. The free tier should give enough real value (full versions with explanations) to build trust, while the paid tier unlocks *volume and analytics* (all versions, topic history), which is the "limited-feature" strategy discussed in [14]. Conversion is a key pilot metric (Section 9).

3.7 Official and industry sources

- The NTC UNT Informatics test specification [15] defines topics, item types and the number of items. It is the basis for topic tags and the blueprint of each version.
- Informburo's 2026 UNT guide [16] provides the current format (120 items, 140 points, 4 hours).
- Informburo's 2021 review of 10 online preparation platforms [17] gives the market and price context (see Section 4).

3.8 Structured analysis of publications

#	Source (year)	Type	Theme	Method / sample	Key finding	Use in project	Limitation
1	Roediger & Karpicke (2006)	Journal, foundational	A	Lab experiments, students	Testing beats restudy for long-term retention	Core rationale for mock tests	Lab setting, prose recall
2	Dunlosky et al. (2013)	Review, foundational	A	Review of 10 techniques	Practice testing has high utility	Spaced repeated attempts	Not domain-specific
3	Adesope et al. (2017)	Meta-analysis	A	272 effect sizes	Practice tests > restudy; format match boosts effect	UNT-identical format	Mostly higher education
4	Yang et al. (2021)	Meta-analysis	A	222 classroom studies	Quizzing boosts classroom learning	Justifies frequent tests	Heterogeneous designs
5	Agarwal et al. (2021)	Systematic review	A	Applied school studies	Retrieval practice benefits are consistent	School-age target users	Few informatics studies
6	Shute (2008)	Review, foundational	B	Literature review	Elaborated, specific, timely feedback works best	Explanation per item	Older, pre-mobile
7	Wisniewski et al. (2020)	Meta-analysis	B	435 studies	Feedback d ≈ 0.48; high-information feedback strongest	Topic and process-level feedback	Wide variance
8	Sailer & Homner (2020)	Meta-analysis	C	Gamification studies	Small positive effects; unstable in rigorous designs	Lightweight progress only	Mixed quality of studies
9	Abdrasilov et al. (2026)	Journal	D	Psychometric validation of UNT	UNT is valid and predicts university grades	High-stakes context	Mathematics only
10	Mingisheva (2023)	Journal	D	UNT data 2014/2017/2020	Urban–rural and language gaps	Kazakh content, online access	Aggregate data
11	Useinov et al. (2022)	Journal	D	Small schools, North Kazakhstan	Online test platforms + guidance improve UNT prep	Online channel for rural students	Small regional sample
12	Rakhmetov et al. (2024)	Journal	D	149 informatics students; teacher survey	Teachers positive but lack platform skills	Teacher segment	University-level sample
13	Gu et al. (2018)	Journal	E	Randomized field experiment, freemium publisher	Premium tiers lift paid sales despite the zero-price pull of free	Tier design (monthly vs. season pass)	Not edtech
14	Tyrväinen & Karjaluoto (2024)	Meta-analysis + interviews	E	55 studies, 15 interviews	Value types → willingness to pay, via trust	Free-tier value, limited-feature model	Mostly apps and games

#	Source (year)	Type	Theme	Method / sample	Key finding	Use in project	Limitation
15	NTC UNT Informatics specification	Official document	D	—	Topics, item types, counts	Item blueprint and topic tags	Changes yearly
16	Informburo (2026) UNT guide	Media	D	—	2026 format facts	Format constraints	Secondary source
17	Informburo (2021) platform review	Media	Market	10 platforms	Prices and features	Competitor context	2021 prices

3.9 What the review means for the project

1. **Approaches that exist:** general test banks, video/live courses and the free official trial. Research strongly supports the *practice test + elaborated feedback* approach.
2. **What is already known:** testing and feedback improve learning; the UNT is high-stakes and valid; there are equity gaps in Kazakhstan; freemium conversion depends on the value of the free tier.
3. **Limitations the project takes into account:** no evidence specific to Informatics or Kazakhstan, so the project measures outcomes in its own pilot. Gamification is kept minimal. Feedback costs content effort, so explanations stay short and written by the author.

References

[1] Roediger, H. L., & Karpicke, J. D. (2006). Test-enhanced learning: Taking memory tests improves long-term retention. *Psychological Science, 17*(3), 249–255. https://doi.org/10.1111/j.1467-9280.2006.01693.x

[2] Dunlosky, J., Rawson, K. A., Marsh, E. J., Nathan, M. J., & Willingham, D. T. (2013). Improving students' learning with effective learning techniques. *Psychological Science in the Public Interest, 14*(1), 4–58. https://doi.org/10.1177/1529100612453266

[3] Adesope, O. O., Trevisan, D. A., & Sundararajan, N. (2017). Rethinking the use of tests: A meta-analysis of practice testing. *Review of Educational Research, 87*(3), 659–701. https://doi.org/10.3102/0034654316689306

[4] Yang, C., Luo, L., Vadillo, M. A., Yu, R., & Shanks, D. R. (2021). Testing (quizzing) boosts classroom learning: A systematic and meta-analytic review. *Psychological Bulletin, 147*(4), 399–435. https://doi.org/10.1037/bul0000309

[5] Agarwal, P. K., Nunes, L. D., & Blunt, J. R. (2021). Retrieval practice consistently benefits student learning: A systematic review of applied research in schools and classrooms. *Educational Psychology Review, 33*, 1409–1453. https://doi.org/10.1007/s10648-021-09595-9

[6] Shute, V. J. (2008). Focus on formative feedback. *Review of Educational Research, 78*(1), 153–189. https://doi.org/10.3102/0034654307313795

[7] Wisniewski, B., Zierer, K., & Hattie, J. (2020). The power of feedback revisited: A meta-analysis of educational feedback research. *Frontiers in Psychology, 10*, 3087. https://doi.org/10.3389/fpsyg.2019.03087

[8] Sailer, M., & Homner, L. (2020). The gamification of learning: A meta-analysis. *Educational Psychology Review, 32*, 77–112. https://doi.org/10.1007/s10648-019-09498-w

[9] Abdrasilov, B. S., Shinetova, L. E., Ivanova, A. E., Kardanova, E. Yu., Beisenova, Z. Zh., Zhabayeva, G. K., Orlov, D., & Ashirbek, G. K. (2026). Validation of the Unified National Testing in Kazakhstan: Evidence-based framework for fair testing. *Voprosy Obrazovaniya / Educational Studies Moscow*, (2), 8–41. https://doi.org/10.17323/vo-2026-27950

[10] Mingisheva, N. A. (2023). Development and challenges of standardized testing in Kazakhstan: Transition from national to international standards. *Journal of Educational Sciences, 76*(3), 94–103. https://doi.org/10.26577/JES.2023.v76.i3.08

[11] Useinov, B. M., Solodovnik, A. A., Shaimerdenova, M. A., & Baizhigitova, L. O. (2022). Features of preparation for the Unified National Testing in small schools within the framework of distance learning. *Bulletin of Abai KazNPU, Series of Physical and Mathematical Sciences, 78*(2), 260–267. https://doi.org/10.51889/2022-2.1728-7901.32

[12] Rakhmetov, M., Kuanbayeva, B., Saltanova, G., Zhusupkalieva, G., & Abdykerimova, E. (2024). Improving the training on creating a distance learning platform in higher education: Evaluating their results. *Frontiers in Education, 9*, 1372002. https://doi.org/10.3389/feduc.2024.1372002

[13] Gu, X., Kannan, P. K., & Ma, L. (2018). Selling the premium in freemium. *Journal of Marketing, 82*(6), 10–27. https://doi.org/10.1177/0022242918807170

[14] Tyrväinen, O., & Karjaluoto, H. (2024). Willingness to pay for freemium services: Addressing the differences between monetization strategies. *International Journal of Information Management, 77*, 102787. https://doi.org/10.1016/j.ijinfomgt.2024.102787

[15] National Testing Center of the Republic of Kazakhstan. *UNT test specification: Informatics.* https://testcenter.kz

[16] Informburo.kz. (2026). *Всё, что нужно знать об ЕНТ-2026: от регистрации до апелляции.* https://informburo.kz/cards/vse-cto-nuzno-znat-ob-ent-2026-ot-registracii-do-apelliacii

[17] Informburo.kz. (2021, November 5). *ҰБТ-ға үйден шықпай-ақ дайындалуға болады: түлектерге арналған 10 онлайн-платформа.* https://informburo.kz/kaz/tarih/ubt-ga-uiden-syqpai-aq-daiyndaluga-bolady-tulekterge-arnalgan-10-onlain-platforma

4. Existing System Comparison

4.1 Selection of systems

Five systems that Kazakhstani students actually use for UNT preparation were selected from the Informburo market review [17] and an October 2026 check of their websites and app-store pages. Together they represent the main approaches on the market:

#	System	Approach	Evidence checked
S1	**NTC trial test** (prob-ent.testcenter.kz)	Official free trial	NTC / news reports (Newtimes.kz, 2022)
S2	**iTest** (itest.kz)	Multi-subject test bank, subscription	Website, Oct 2026
S3	**Joo.kz**	Course + weekly mock tests + app	Website, Oct 2026
S4	**Daryn.online**	Video/live course platform	Website, Oct 2026
S5	**Baimly**	Mobile app with lessons and UNT-format tests	App Store page, Oct 2026

4.2 Comparison criteria

Code	Criterion	Why it matters (link to evidence)
C1	Informatics focus and depth	Core gap (Section 2, finding 3)
C2	Full UNT-format mock test (timed, real structure)	Format match improves transfer [3]
C3	Item-level explanations	Elaborated feedback [6], [7]
C4	Topic-level analytics of weaknesses	High-information feedback [7]
C5	Kazakh-language content	Language gap [10]
C6	Modern item types (Python code, SQL tables)	Current UNT Informatics format (finding 4)
C7	Free tier available	Trust and value before paying [14]
C8	Price of paid access	Price sensitivity (assumption to test)

4.3 Comparison table

Legend: ✔ = yes, confirmed; ◐ = partial; ✘ = no; **n/s** = not stated publicly (could not be verified).

Criterion	S1 NTC trial	S2 iTest	S3 Joo.kz	S4 Daryn.online	S5 Baimly	Proposed platform
C1 Informatics focus	◐ one of many subjects	◐ one of 20 subjects	◐ mainly the Mathematics–Informatics pair; no separate Informatics course described	✘ Informatics not named as a subject	n/s	✔ Informatics only
C2 Full UNT mock	✔ official format	✔ “Пробный ЕНТ”	✔ weekly 120-item, 4-hour mock	✔ weekly trial	◐ UNT-format tests	✔ 40-item Informatics block, timed
C3 Item explanations	n/s	◐ study notes; explanations not stated	✔ answers + explanations	n/s	✔ video analysis per lesson	✔ text explanation per item

Criterion	S1 NTC trial	S2 iTest	S3 Joo.kz	S4 Daryn.online	S5 Baimly	Proposed platform
C4 Topic analytics	✔ thematic analysis	✔ “detailed statistics”	◐ leaderboard; AI tutor	✔ diagnostics by section	n/s	✔ per-topic mastery + history
C5 Kazakh	✔	✔ KZ/RU	✔ KZ/RU	✔ KZ	✔ KZ	✔ KZ (RU later)
C6 Code/SQL items	n/s	n/s	n/s	n/s	n/s	✔ highlighted code, SQL tables
C7 Free tier	✔ fully free	◐ limited	✔ free weekly mock	✔ free weekly trial	✔ free app (paid courses n/s)	✔ free versions with full explanations
C8 Price	Free	4,490 KZT / 90 days; 14,290 KZT / year	Course price not listed (2021: 84,000–99,600 KZT)	Not listed (2021: 9,900–69,000 KZT)	n/s	Planned: 1,990 KZT/month or 7,990 KZT until UNT (to validate)

Sources: S1 – Newtimes.kz (2022), Informburo [17]; S2–S4 – official websites accessed 21 Oct 2026 and [17]; S5 – App Store listing (v3.9.0, July 2025, rating 3.5 from 2 reviews).

4.4 Analysis

- **The market is crowded for general UNT preparation but thin for Informatics.** None of the five systems describes a dedicated, deep Informatics product. The NTC trial is free and official, so it is a strong baseline for *format*, but no public source shows that it explains each item.
- **Claims about modern item types could not be verified** for any competitor (C6 = n/s). This is noted as an *assumption* about competitor weakness rather than a fact, and it will be checked during the pilot by asking users to compare.
- **Price:** iTest shows that students accept cheap subscriptions (about 1,500 KZT per month). Course platforms charge far more. The proposed price sits at the test-bank level, not the course level.
- **Free official alternative:** because the NTC trial is free, the platform must win on *explanations, Informatics depth and repeated practice*, not on format alone.

4.5 Why the proposed solution is needed

The proposed platform does not compete with full courses. It fills a niche: **an Informatics-only, Kazakh-first practice-test service with item-level explanations and topic analytics, at test-bank prices.** This is supported by research on retrieval practice and elaborated feedback (Section 3) and by the gaps in table 4.3. The weakest part of the argument, user willingness to pay, is treated as a hypothesis and tested in the pilot (Section 9).

5. Technology Comparison

Each decision is scored against criteria derived from the requirements (Section 6) and the project constraints: one developer, a short timeline, a low budget, and a need for server-side rendering and SEO for marketing pages. Scores are 1 (poor) to 5 (best); weights add up to 100%.

5.1 Decision 1: Web application framework (frontend + server)

Criterion (weight)	Next.js (React, App Router)	Django + templates/HTMX	Separate React SPA + Express API
Interactive test runner UI with timer and state (25%)	5	3	5
SEO and fast first load for landing/pricing pages (15%)	5 (SSR/SSG)	4	2 (client-only rendering)
Developer productivity for one person, single codebase (20%)	5 (full-stack TypeScript)	4	3 (two apps to maintain)
Ecosystem: auth, payments, UI kits (15%)	5	4	4
Hosting cost and simplicity (15%)	4 (Vercel/Render free tier)	3	3
Author’s existing skills (10%)	3	4 (Python)	3
Weighted score	**4.65**	3.60	3.50

Decision: Next.js with TypeScript. It gives one codebase for UI and API (Route Handlers and Server Actions), SSR for marketing pages and a rich interactive runner. **Trade-off:** less familiar to the author than Python, which is mitigated by a learning spike in Sprint 1 (YouTrack UNT-6).

5.2 Decision 2: Data storage

Criterion (weight)	PostgreSQL	MongoDB	Firebase Firestore
Relational integrity: users–subscriptions–attempts–answers (30%)	5	3	2
Analytical queries (topic mastery, conversion funnels) in SQL (20%)	5	3	2
Flexible item content (JSON for code/table blocks) (15%)	4 (JSONB)	5	4
Transactions for payment and entitlement updates (15%)	5	4	3
Free/managed hosting (Neon, Supabase, Render) (10%)	5	4	4
Vendor lock-in risk (10%)	5 (open standard)	4	2
Weighted score	**4.85**	3.65	2.65

Decision: PostgreSQL with Prisma ORM for migrations and type-safe queries. Item bodies with code or table blocks are stored as validated JSONB. **Trade-off:** a schema change needs a migration, which is acceptable and also gives a traceable history.

5.3 Decision 3: Component interaction and API style

Criterion (weight)	Server Actions + REST route handlers (monolith)	GraphQL API	Microservices (separate scoring/payment services)
Simplicity for MVP and one developer (35%)	5	3	1

Criterion (weight)	Server Actions + REST route handlers (monolith)	GraphQL API	Microservices (separate scoring/payment services)
Clear contract for webhooks (payment provider) (20%)	5 (REST endpoint)	3	4
Performance for a test session (load once, submit once) (20%)	5	4	3
Testability (15%)	4	4	3
Future scalability (10%)	3	4	5
Weighted score	**4.65**	3.45	2.70

Decision: modular monolith. Internal modules (auth, catalogue, test session, scoring, billing, admin) have clear boundaries (see the component diagram, Section 7). External integrations use REST: the payment webhook and the import endpoint. **Trade-off:** scaling means scaling the whole app, which is acceptable for the expected pilot load (< 500 concurrent users).

5.4 Decision 4: Payments (subscription)

Criterion (weight)	Kaspi Pay / local acquiring (via aggregator, e.g. CloudPayments KZ)	Stripe	Manual (Kaspi transfer + admin activation)
Availability for a Kazakhstani merchant and payers (35%)	5	2 (not directly available to KZ merchants)	5
Recurring payments and webhooks (25%)	4	5	1
Test mode for development (20%)	4	5	1
Fees and setup effort (20%)	3	3	5
Weighted score	**4.15**	3.55	3.20

Decision: integrate a local aggregator's **test mode** during the MVP. Billing sits behind an interface (`PaymentProvider`), so the provider can be switched. Manual activation is the fallback for the first pilot customers. Final choice of provider will be made after the legal registration of the business (risk R5).

5.5 Supporting choices (not weighted)

Area	Choice	Alternatives considered	Reason
Auth	Auth.js (NextAuth) credentials + email verification	Clerk, Firebase Auth	Free, self-hosted, no user data sent to third parties
Hosting	Vercel (app) + Neon/Supabase (PostgreSQL)	Render, VPS (PS Cloud)	Free tiers for the pilot; a VPS in Kazakhstan if data-residency requirements apply
Testing	Vitest (unit), Playwright (E2E)	Jest, Cypress	Fast, TypeScript-native, CI-friendly
CI	GitHub Actions	GitLab CI	Repository is on GitHub
Code highlighting	Shiki	Prism	Server-side rendering, accurate Python grammar
Diagrams	PlantUML (sources in `diagrams/src`)	draw.io	Text-based, versioned in Git

6. Requirements Specification (initial)

Priorities follow **MoSCoW**: **M** = Must (MVP is not accepted without it), **S** = Should, **C** = Could, **W** = Won't (this release). Every *Must* requirement has an acceptance criterion that can be checked by a test (Section 9). "Source" links each requirement to evidence (Section 2) or literature (Section 3).

6.1 Actors

- **Guest:** an unregistered visitor.
- **Student:** a registered user on the free or premium plan.
- **Administrator:** the content owner and operator.
- **Payment provider:** an external system.

6.2 Functional requirements

ID	Requirement	Priority	Source	Acceptance criteria
FR-01	A guest can register with email and password and verify the email	M	—	Given a valid email and password (≥ 8 characters), an account is created and a verification link is sent. An unverified user cannot start a test. A duplicate email is rejected with a message.
FR-02	A student can log in, log out and reset a forgotten password	M	—	A valid login creates a session. An invalid login shows an error after ≤ 1 s. A reset link expires after 60 minutes.
FR-03	A guest/student can browse the catalogue of Informatics test versions with a free/premium label	M	E2	The catalogue lists all *published* versions with number, difficulty and lock icon. Unpublished versions are never shown.
FR-04	A guest can try one demo version without registering	S	[14]	The demo opens without login. The results page asks the guest to register to save them.
FR-05	A student can take a timed test that copies the UNT Informatics block (40 items, single choice, multiple choice, matching and context-block items)	M	E1, [3]	The timer counts down from the configured limit. The student can move between items and flag them. Answers are saved within 2 s of selection. A reload or reconnect restores the session with the remaining time. The test auto-submits at 00:00.
FR-06	The system scores a submitted attempt according to UNT rules	M	E1, [15]	Single-choice items: 1/0. Multiple-choice and matching items: partial credit per the specification table. The total equals the sum of item scores. Scoring unit tests cover every item type and edge case (no answer, all options selected).
FR-07	Items can contain formatted Python code, SQL tables, images and spreadsheet fragments	M	Finding 4	Code renders with syntax highlighting and preserved indentation. Tables render with headers. Images have alt text. The rendering is identical in the admin preview and the test runner.
FR-08	After submission, the student sees the correct answer and an explanation for every item	M	[6], [7]	Each item on the results page shows the student's answer, the correct answer and a non-empty explanation. Publishing a version with an empty explanation is blocked.
FR-09	The results page shows a breakdown by specification topic	M	[7]	Every item has ≥ 1 topic tag. The results page shows the percentage correct per topic, and the topic with the lowest score is highlighted.
FR-10	A student can view their attempt history and progress chart	S	[2], [5]	The history lists all attempts with date, version and score. The chart shows the score over time when there are ≥ 2 attempts.

ID	Requirement	Priority	Source	Acceptance criteria
FR-11	The free plan gives access to a limited set of versions	M	[14]	Free users can open exactly N versions (configurable, default 3). Opening a premium version shows the paywall. Access is checked on the server, so direct URL access is denied with 403.
FR-12	A student can buy a subscription (monthly or “until UNT”)	M	[13]	Checkout redirects to the provider (test mode). After a successful webhook, the plan becomes premium within 1 minute. A failed or cancelled payment leaves the plan unchanged. A webhook signature is verified, and an invalid signature is rejected.
FR-13	Subscription expiry is handled automatically	M	—	When `paid_until` < now, premium versions become locked again. The student sees the expiry date in the profile.
FR-14	The administrator can import a test version from a JSON file (with images) and preview it	M	O5	Import validates the schema (40 items, item types, answer keys, topics, explanations) and lists errors by item number. A valid file creates a *draft* version. The preview shows the version as the student would see it.
FR-15	The administrator can publish/unpublish versions and mark them free/premium	M	—	A status change is visible in the catalogue immediately. All status changes are logged with time and admin.
FR-16	The system records product analytics events (registration, test start/finish, paywall view, checkout start, payment)	M	O6, O7	Each listed event is stored with user ID (pseudonymous), timestamp and context. The admin can export events as CSV for the conversion funnel.
FR-17	The administrator sees a basic dashboard (users, attempts, conversion)	S	O7	The dashboard shows counts for the last 7/30 days, matching SQL checks on the events table.
FR-18	A student can report an error in an item	S	Quality	A “report” button saves the item ID and comment. The admin sees the list of open reports.
FR-19	Russian-language interface and content	C	[10]	All UI strings are available in RU. Content is shown in RU when a translation exists.
FR-20	Teacher accounts that assign tests to a class	C	[12]	Future release.
FR-21	Adaptive test generation / AI explanations	W	—	Out of scope (future work).

6.3 Non-functional requirements

ID	Category	Requirement	Priority	Acceptance criteria / measurement
NFR-01	Performance	Fast page loads on mobile networks	M	Lighthouse Performance ≥ 80 on mobile for the catalogue and runner pages. p95 server response for saving an answer ≤ 300 ms at 100 concurrent sessions (k6 load test).
NFR-02	Reliability	No loss of answers during a test	M	In an E2E test, killing the network for 30 s and then restoring it keeps all answers, and the attempt can still be submitted.
NFR-03	Security	Protection against the OWASP Top 10 basics	M	Passwords hashed with bcrypt/argon2. All forms CSRF-protected. Authorization checked server-side for every premium resource (automated test). No secrets in the repository (gitleaks in CI passes).
NFR-04	Security	Answer keys are never sent to the client before submission	M	Inspecting network responses during a test shows no `correct` fields (automated E2E check).

ID	Category	Requirement	Priority	Acceptance criteria / measurement
NFR-05	Privacy	Minimal personal data; compliance with the Law of RK “On Personal Data and Their Protection”	M	Only email, display name and grade are collected. A privacy policy page exists. A user can delete their account, and the data is removed within 30 days.
NFR-06	Usability	The platform is easy to use for school students	M	SUS score ≥ 70 in the pilot (n ≥ 10). A new user completes registration and starts a test in ≤ 3 minutes without help.
NFR-07	Localization	Kazakh interface and content first	M	100% of UI strings in Kazakh. Language is switched through i18n files, with no hard-coded strings (lint rule).
NFR-08	Compatibility	Works on current mobile and desktop browsers	M	E2E tests pass on Chromium, WebKit and a mobile viewport of 375 px.
NFR-09	Accessibility	Basic WCAG 2.1 AA	S	axe-core shows no critical violations on the main pages.
NFR-10	Maintainability	Automated tests and CI	M	CI runs lint, unit and E2E tests on every pull request. Scoring and billing modules have ≥ 80% line coverage.
NFR-11	Availability	Uptime during the pilot	S	≥ 99% uptime measured by an external monitor over the pilot period.
NFR-12	Cost	Low operating cost	S	Hosting cost during the pilot ≤ 10,000 KZT per month.

6.4 Constraints

- Item content must be original. It is written to the public NTC specification, and NTC items are not copied.
- Payments run in test mode until the business is registered.
- Pilot participants under 18 take part only with consent, and no unnecessary personal data is collected.

6.5 Traceability summary

Objective	Requirements
O4 Build MVP	FR-01…FR-15, NFR-01…NFR-10
O5 Content	FR-07, FR-14, FR-18
O6 Pilot evaluation	FR-09, FR-10, FR-16, NFR-06
O7 Commercialization	FR-11…FR-13, FR-16, FR-17

7. Architecture and Diagrams

Editable sources of all diagrams are in `diagrams/src/*.puml` (PlantUML) and rendered images are in `diagrams/png/`. To regenerate them:

```
java -jar plantuml.jar -tpng -o ../png diagrams/src/*.puml
```

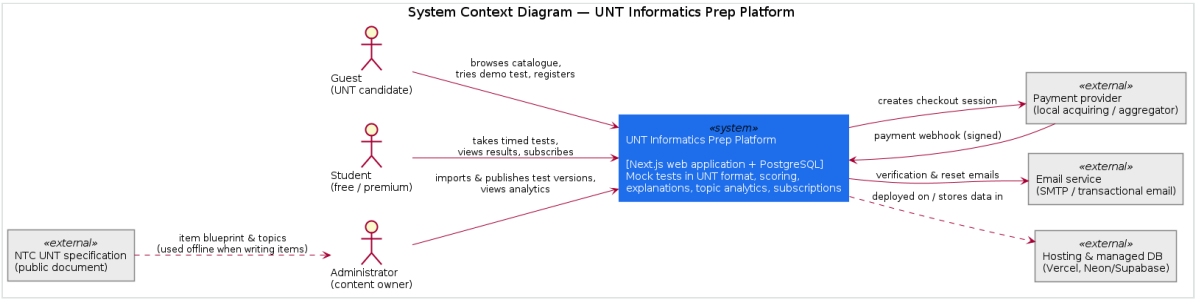
7.1 Architectural overview

The platform is a **modular monolith** built with Next.js (TypeScript), using a PostgreSQL database through the Prisma ORM. It is deployed on Vercel with a managed PostgreSQL instance (Neon or Supabase). The decisions behind this choice are in Section 5.

Key architectural principles:

1. **Server-side authority:** entitlement checks, scoring and answer keys live only on the server. The client receives items without keys (NFR-04).
2. **Clear module boundaries** inside one deployable unit: Auth, Catalogue, Test Session, Scoring, Results & Analytics, Billing, Content Admin and Event tracker. Each module accesses the database through a shared data-access layer.
3. **Replaceable external providers:** payments sit behind a `PaymentProvider` interface and email behind a `Mailer` interface.
4. **Resumable test sessions:** an `Attempt` row stores `deadline_at`, and each answer is upserted immediately. The timer is computed from the server deadline, so a reload or network loss does not reset it (FR-05, NFR-02).

7.2 System context diagram

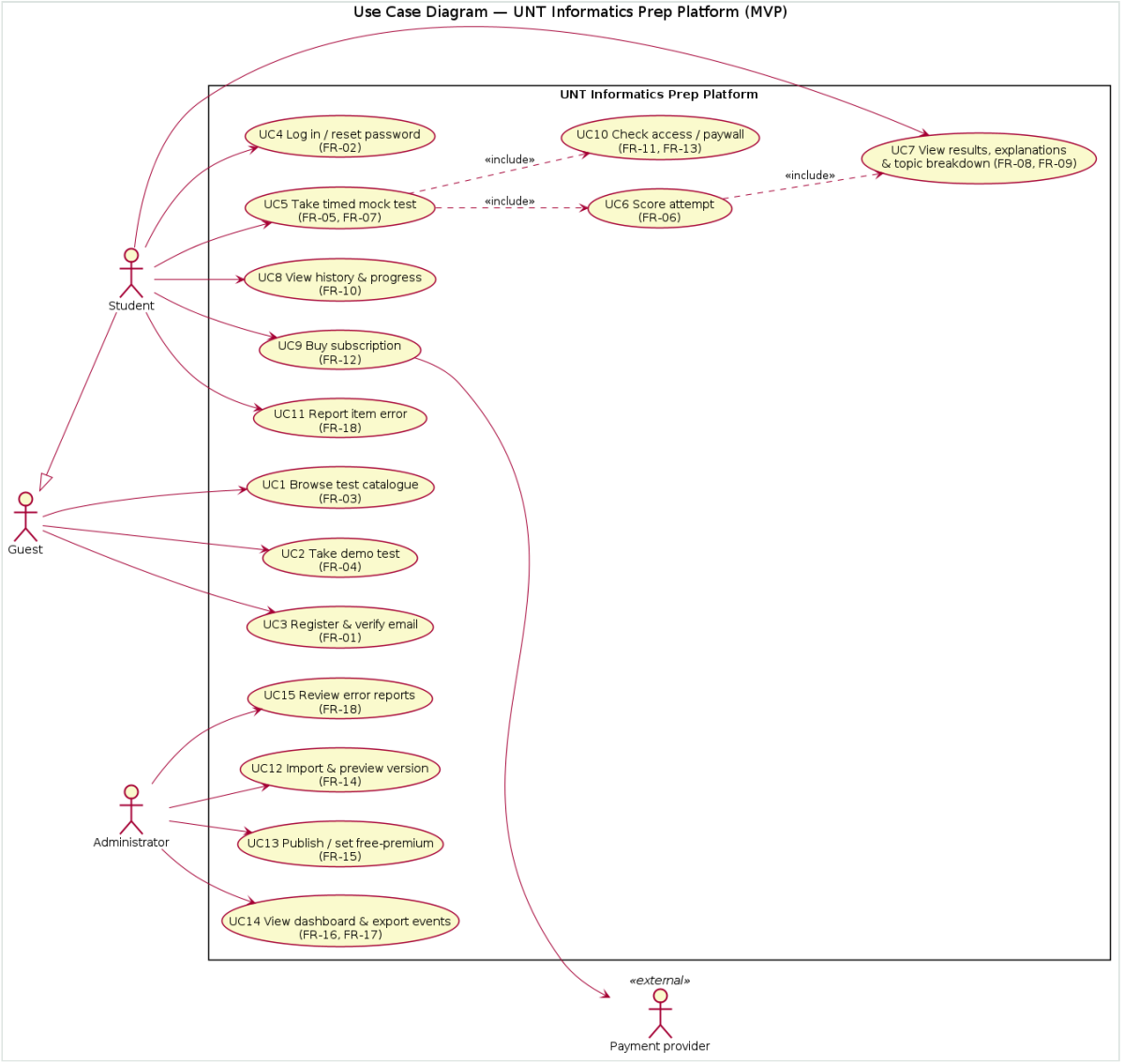


System context

Source: `diagrams/src/01-system-context.puml`

The platform interacts with three human actors (guest, student, administrator) and with three external systems: the payment provider (checkout and signed webhook), the email service (verification and password reset) and the hosting/database provider. The NTC specification is used offline by the administrator when writing items.

7.3 Use case diagram

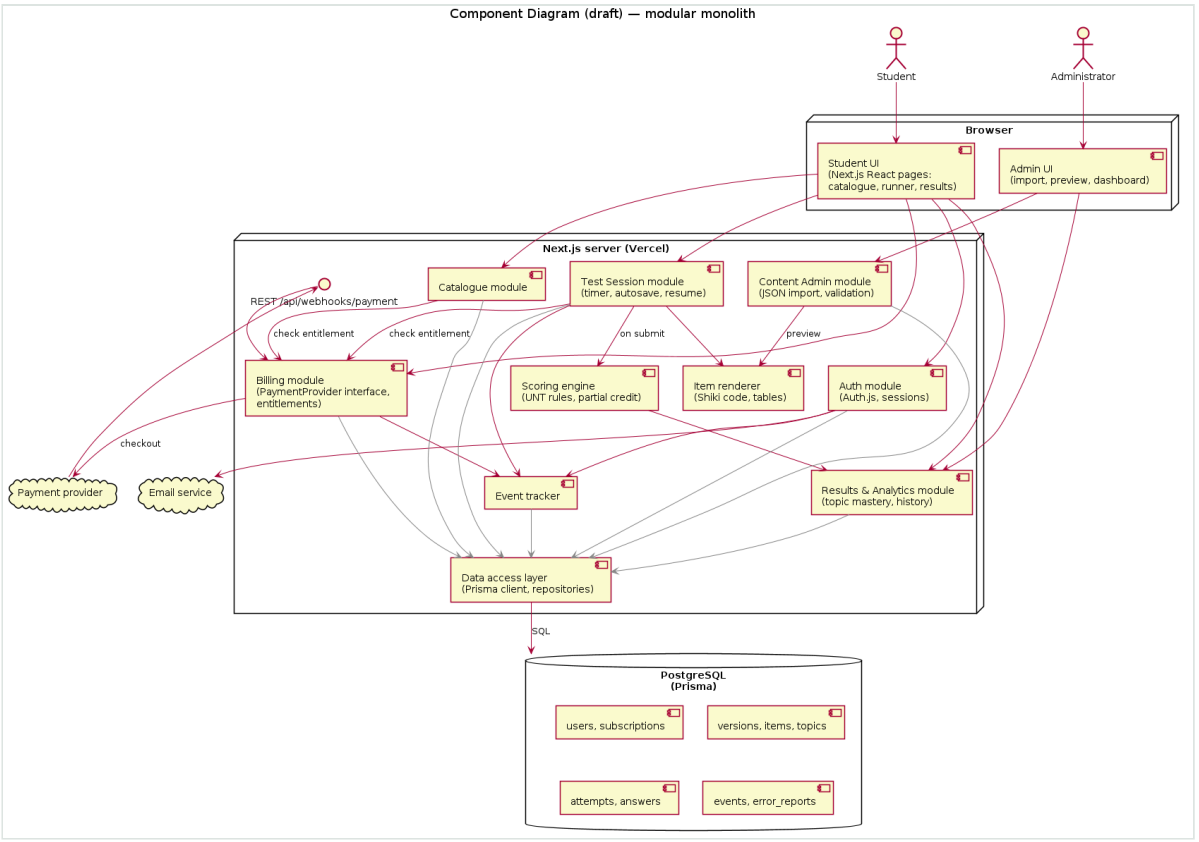


Use case

Source: `diagrams/src/02-use-case.puml`

Every use case carries the IDs of the functional requirements it covers (Section 6), so each *Must* requirement can be traced to at least one use case.

7.4 Component diagram (draft)

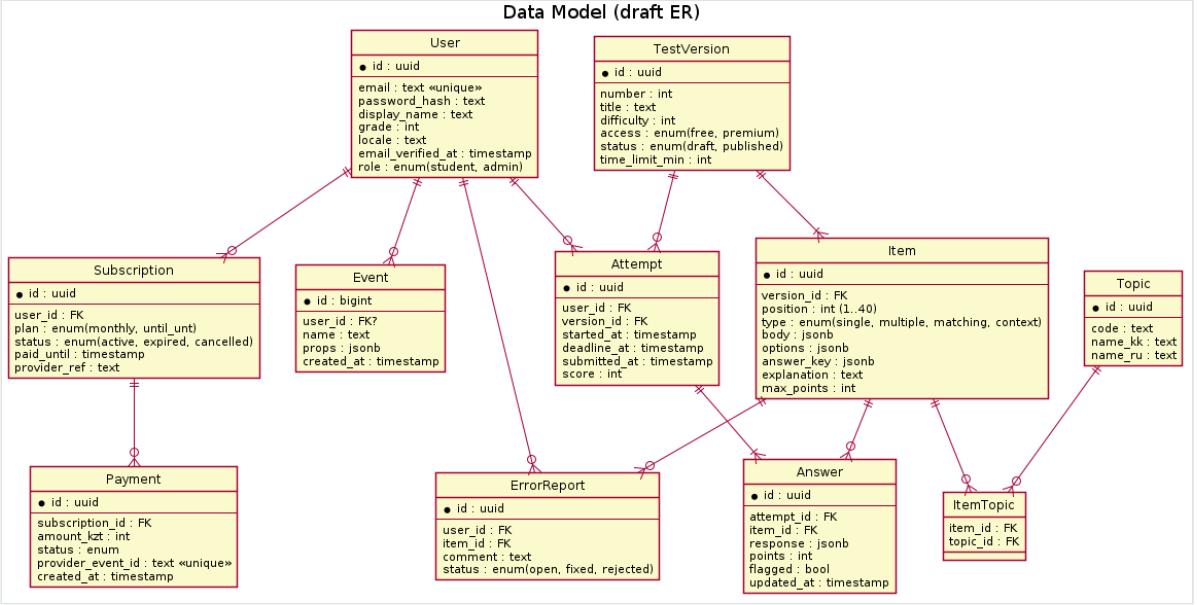


Component

Source: `diagrams/src/03-component.puml`

Component	Responsibility	Requirements
Student UI / Admin UI	React pages and forms; runner timer display; no answer keys	FR-03…FR-10, FR-14…FR-17
Auth module	Registration, verification, login and sessions, roles	FR-01, FR-02, NFR-03
Catalogue module	Lists published versions and access labels	FR-03, FR-11
Test Session module	Starts an attempt, autosaves answers, resumes, auto-submits	FR-05, NFR-02
Item renderer	Renders code (Shiki), tables, images in the same way for student and admin	FR-07
Scoring engine	Pure functions implementing UNT scoring per item type	FR-06
Results & Analytics	Explanations, topic mastery, history, admin dashboard	FR-08…FR-10, FR-17
Billing module	Checkout, webhook verification, entitlements, expiry	FR-11…FR-13
Content Admin module	JSON import, schema validation, preview, publish	FR-14, FR-15, FR-18
Event tracker	Records product events for funnels	FR-16
Data access layer	Prisma client and repositories; transactions	all

7.5 Data model (draft)



Data model

Source: `diagrams/src/04-data-model.puml`

`Item.body`, `Item.options` and `Item.answer_key` are JSONB so that the four UNT item types and content blocks (code, table, image) fit one table. A JSON Schema validates them on import.

7.6 Key scenario: taking a test (sequence, textual)

1. The student opens a version, and the **Catalogue** asks **Billing** whether the user is entitled to it. If not, the paywall is shown (FR-11).
2. **Test Session** creates an `Attempt` with `deadline_at = now + time_limit` and returns the items *without* answer keys (NFR-04).
3. Every answer change is upserted into `Answer` (FR-05). On reload, the session is restored from the database.
4. On submit or at the deadline, the **Scoring engine** computes points per item, and the `Attempt.score` is stored (FR-06).
5. **Results & Analytics** returns explanations and topic mastery (FR-08, FR-09). The **Event tracker** logs `test_finished`.

7.7 Alignment check

Requirement group	Use cases	Components	Data
Account (FR-01, FR-02)	UC3, UC4	Auth	User
Testing (FR-05…FR-07)	UC5, UC6	Test Session, Scoring, Renderer	Attempt, Answer, Item
Feedback (FR-08…FR-10)	UC7, UC8	Results & Analytics	Answer, ItemTopic, Topic
Monetization (FR-11…FR-13)	UC9, UC10	Billing	Subscription, Payment
Content (FR-14, FR-15, FR-18)	UC11–UC13, UC15	Content Admin	TestVersion, Item, ErrorReport
Analytics (FR-16, FR-17)	UC14	Event tracker, Results & Analytics	Event

8. UX / Prototype

8.1 Prototype links

- **Clickable prototype (HTML):** `prototype/index.html` in the repository. Open it in any browser; it needs no installation.
- **Online version:** https://yerrrzat.github.io/Web-platform-UNT-/prototype/ (GitHub Pages, public)
- **Screenshots:** `prototype/screenshots/`

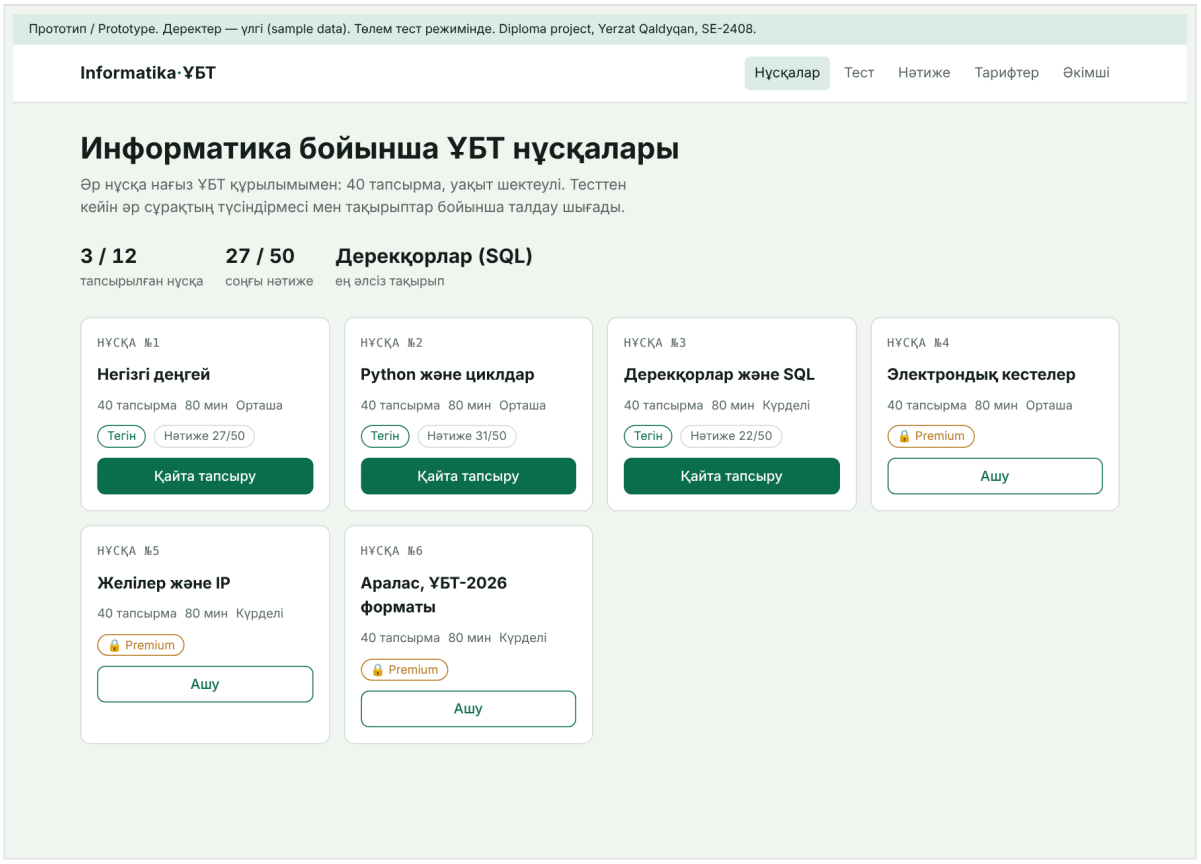
The prototype is in Kazakh (the primary interface language, NFR-07) and uses sample data. It is interactive: the timer runs, answers are selected and flagged, the test can be submitted, and the score, explanations and topic breakdown are calculated with the real scoring rules (FR-06).

8.2 Key user scenarios

#	Scenario	Screens	Requirements
S1	A student chooses a free version and starts a test	Catalogue → Test	FR-03, FR-05, FR-11
S2	A student answers items with code and SQL tables, flags an item, navigates through the answer sheet and submits	Test	FR-05, FR-07
S3	A student reviews the score, the weakest topic and the explanation for each item	Results	FR-06, FR-08, FR-09
S4	A student opens a premium version and is offered a subscription	Catalogue → Pricing	FR-11, FR-12
S5	The administrator imports a version, sees validation errors and cannot publish until they are fixed	Admin	FR-14, FR-15, FR-08
S6	The administrator checks the registration-to-payment funnel	Admin	FR-16, FR-17

8.3 Screens

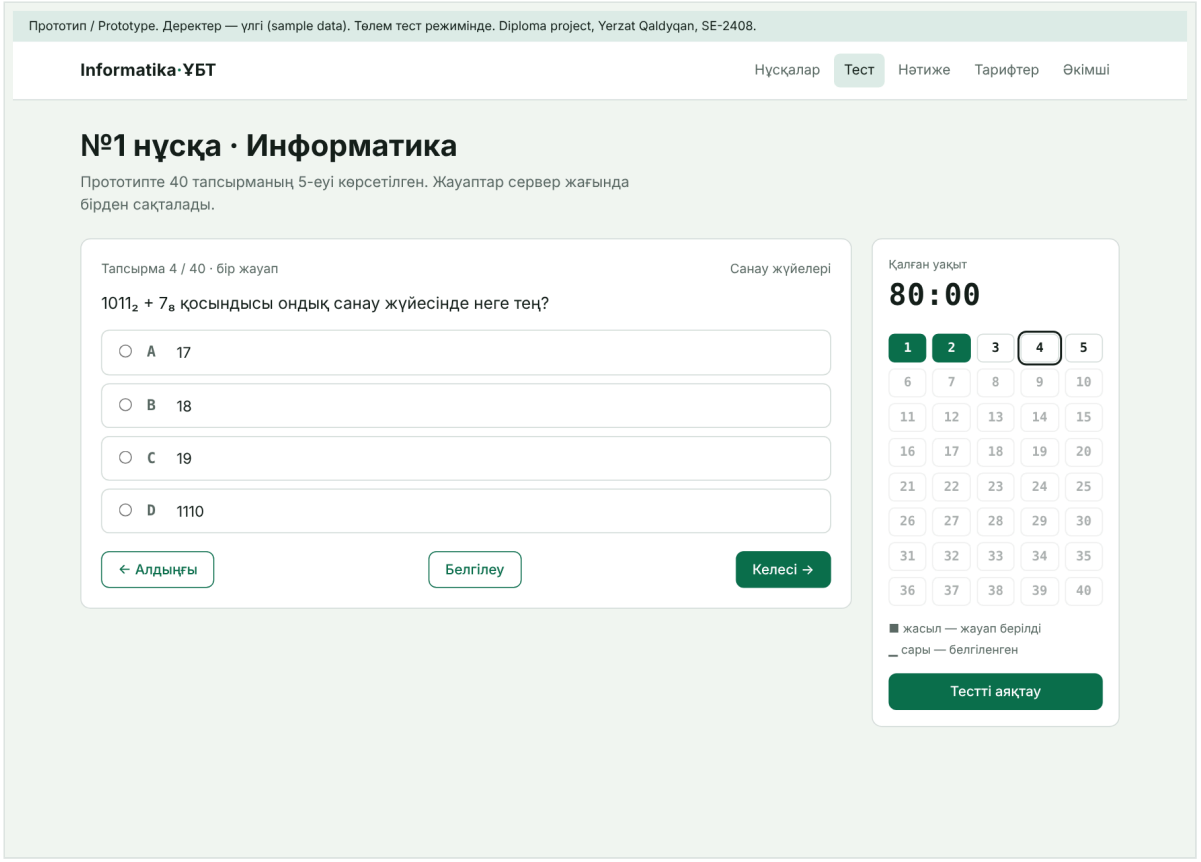
Catalogue (S1, S4)



Catalogue

Versions show the number, theme, difficulty, a free/premium label and the student’s last result. Premium versions lead to the pricing page instead of the test.

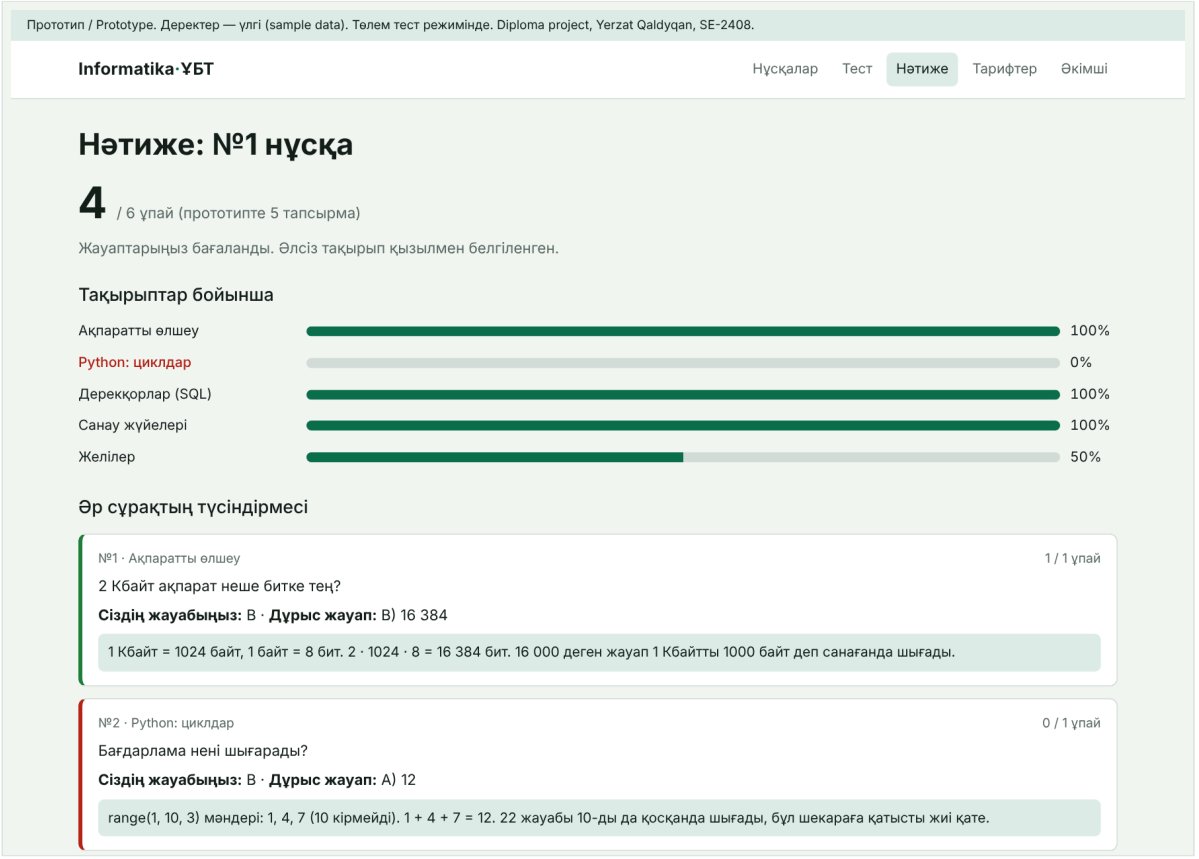
Test runner (S2)



Test runner

The layout follows the UNT computer-based interface: one item at a time, an answer sheet of 40 cells on the side, a countdown timer that turns red under 5 minutes, flagging, and previous/next navigation. Code is shown in a monospaced block with highlighting; SQL items show the table above the query.

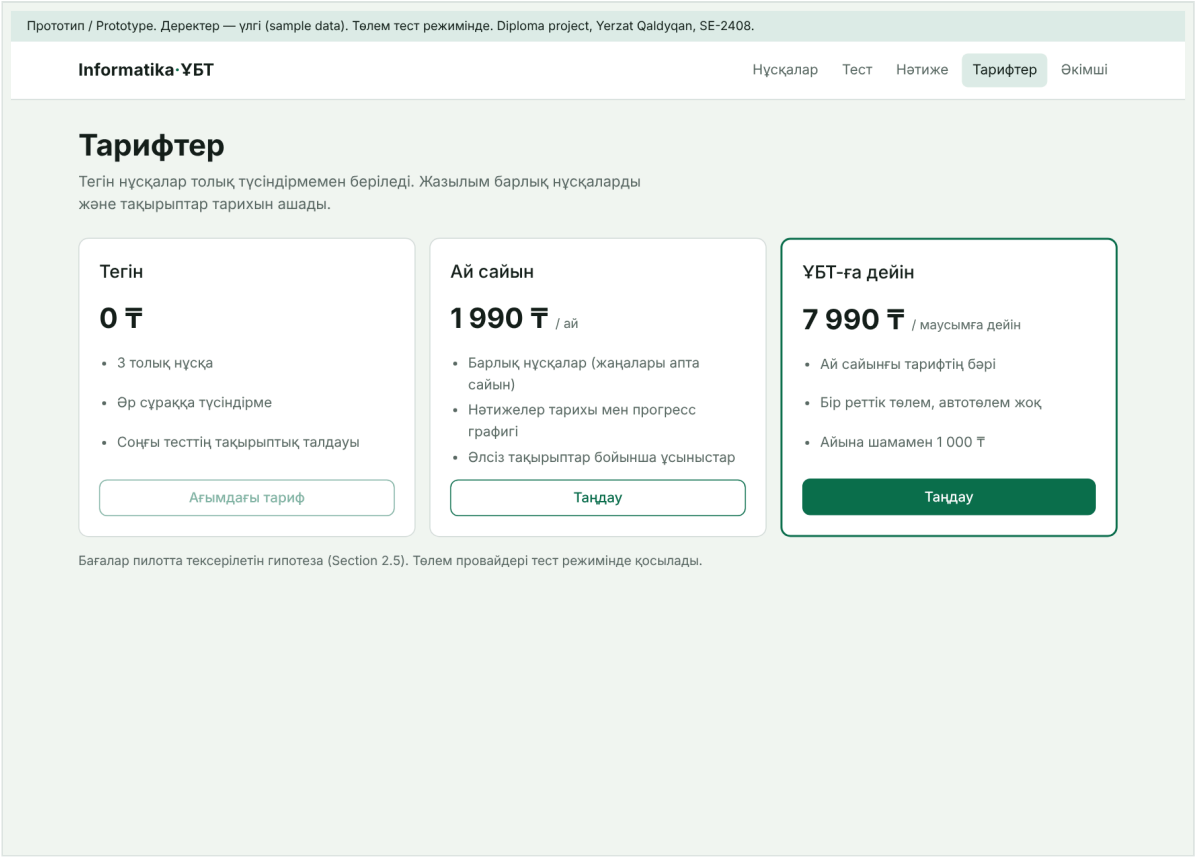
Results (S3)



Results

The results page shows the total score, a bar per specification topic (the weakest is highlighted in red) and every item with the student’s answer, the correct answer and an explanation that names the typical error.

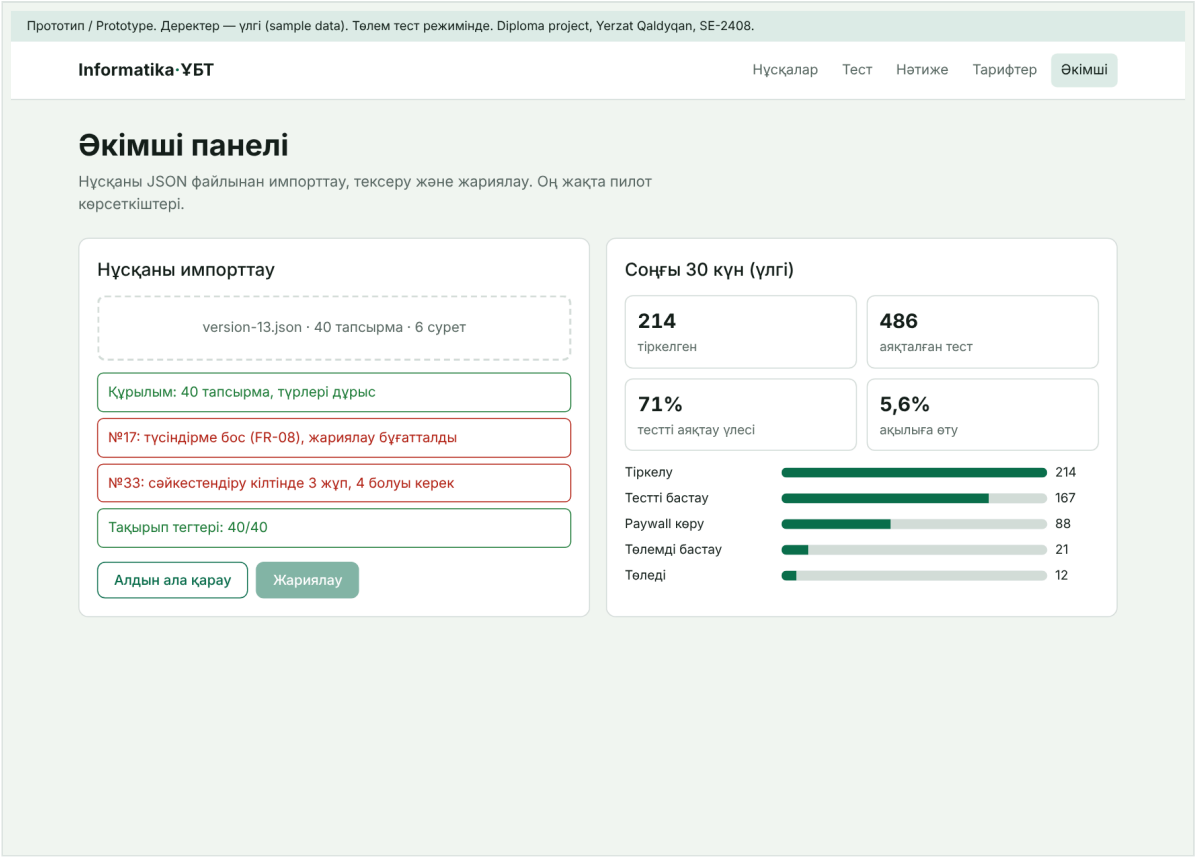
Pricing (S4)



Pricing

Three plans: Free (3 versions with full explanations), Monthly (1,990 KZT) and Until UNT (7,990 KZT, one payment). The second premium tier is a deliberate design choice based on Gu et al. (2018) [13]. Prices are hypotheses to be tested in the pilot.

Admin (S5, S6)



Admin

Import validation lists errors per item number. Publishing is blocked while an explanation is missing. The dashboard shows the registration → test → paywall → checkout → payment funnel used for the commercialization assessment.

8.4 Example input data: version import format (JSON)

```
        {
          "number": 13,
          "title": "Аралас, ҰБТ-2026 форматы",
          "timeLimitMin": 80,
          "access": "premium",
          "items": [
            {
              "position": 2,
              "type": "single",
              "topics": ["python-loops"],
              "body": [
                {"kind": "text", "value": "Бағдарлама нені шығарады?"},
                {"kind": "code", "lang": "python", "value": "s = 0\nfor i in range(1, 10, 3):\n
s += i\nprint(s)"}
              ],
              "options": ["12", "22", "15", "10"],
              "answerKey": [0],
              "maxPoints": 1,
              "explanation": "range(1, 10, 3) мәндері: 1, 4, 7. 1 + 4 + 7 = 12."
            }
          ]
        }
```

8.5 Planned usability checks

- Hallway test of the prototype with 3 students before development starts (Sprint 2). Task success and time are recorded for scenarios S1–S3.
- SUS questionnaire in the pilot (NFR-06, Section 9).

9. Verification and Evaluation Plan

The plan answers two questions:

- **Verification:** does the MVP meet its requirements?
- **Evaluation:** does it achieve the project aim, meaning it helps students practise effectively and shows commercial potential?

9.1 Verification (testing plan)

Level	Tool	What is tested	Requirements	When
Unit	Vitest	Scoring engine (every item type, partial credit, empty answers), entitlement rules, import schema validation, subscription expiry	FR-06, FR-11, FR-13, FR-14	Every PR (CI)
Integration	Vitest + test PostgreSQL (Docker)	Attempt lifecycle (start, save, resume, submit), payment webhook with valid/invalid signature, event recording	FR-05, FR-12, FR-16	Every PR (CI)
End-to-end	Playwright (Chromium, WebKit, 375 px mobile)	Scenarios S1–S6 from Section 8; network drop during a test; no answer keys in network responses	FR-01…FR-15, NFR-02, NFR-04, NFR-08	Every PR to `main`
Performance	k6	100 concurrent sessions saving answers; p95 ≤ 300 ms	NFR-01	Before pilot
Front-end quality	Lighthouse CI, axe-core	Performance ≥ 80 (mobile), no critical accessibility issues	NFR-01, NFR-09	Weekly / before release
Security	gitleaks, npm audit, manual OWASP checklist	No secrets, no known critical vulnerabilities, authorization on every premium route	NFR-03	Every PR + before pilot
Content QA	Import validator + peer review of 1 version by an informatics teacher	No empty explanations, correct keys, topic tags	FR-08, FR-14	Each new version

Acceptance rule: a *Must* requirement is accepted only when its acceptance criterion (Section 6) is covered by a passing automated test or by a recorded manual check in the test log (`tests/README.md`).

9.2 Evaluation: outcome metrics

Metric	Definition	Target	Data source
M1 Activation	% of registered users who finish ≥ 1 full test	≥ 50%	Events `registered`, `test_finished`
M2 Test completion rate	Finished / started attempts	≥ 70%	Attempts table
M3 Learning progress	Mean change in score (%) between a student's 1st and 3rd attempt on *different* versions of similar difficulty	≥ +10 percentage points	Attempts, Answers
M4 Weak-topic improvement	Change in % correct in the student's weakest topic after ≥ 2 more attempts	Positive, significant (paired t-test / Wilcoxon, α = 0.05)	Answers + ItemTopic
M5 Usability	System Usability Scale	≥ 70	Survey (n ≥ 10)

Metric	Definition	Target	Data source
M6 Perceived value of explanations	Share of students rating explanations as “useful” or “very useful”	≥ 70%	Survey
M7 Paywall conversion	Paid users / users who saw the paywall	≥ 3% (benchmark range for freemium is low single digits)	Events
M8 Willingness to pay	Share choosing a paid plan, or saying they would pay at the tested price	≥ 20% of surveyed users	Survey + checkout starts
M9 Reliability	Uptime during the pilot	≥ 99%	External monitor

9.3 Measurement conditions

- **Pilot:** 4 weeks in March 2027 (after development is completed by 28 February 2027, as set in the supervisor’s plan) with **≥ 30 students** on the Mathematics–Informatics profile. Recruited through informatics teachers and the author’s student groups, with consent (parental consent for under-18s). Only pseudonymous IDs are used in the analysis.
- **Devices:** students’ own phones and computers. The device type is logged to compare mobile and desktop use.
- **Content:** ≥ 10 versions of comparable difficulty (the blueprint is checked against the specification), so that score changes reflect learning rather than easier versions.
- **Payments:** test mode, or real payments if the business is registered by then. Without real payments, M7 is measured as “checkout started” and reported as *intent*, not revenue.

9.4 Baseline for comparison

Baseline	How it is used
B1: NTC free trial test (official format, score + thematic analysis)	In the survey, students who used both rate usefulness of feedback (M6) for each. The project claims an advantage only for item-level explanations.
B2: Practice without explanations (internal A/B)	During the pilot, half of the users (random assignment) see only correct answers for their first 2 attempts, and the other half see full explanations. M3 and M4 are compared between groups. This isolates the effect of the key feature.
B3: Market price baseline	iTest (≈ 1,500 KZT per month equivalent) vs. course platforms (tens of thousands of KZT). Used to interpret M7 and M8.

9.5 Commercialization assessment

The pilot data feed the commercialization assessment in the final report:

1. Market size: the number of UNT participants choosing the Mathematics–Informatics pair (NTC statistics) → TAM/SAM/SOM.
2. Unit economics: price, M7 conversion, hosting and payment fees, customer acquisition cost from pilot channels → break-even number of subscribers.
3. A go/no-go recommendation with the conditions for continuing.

9.6 Threats to validity

- *Selection bias:* pilot volunteers are motivated students. This is reported, and results are not generalised to all candidates.
- *Small sample:* effect sizes and confidence intervals are reported, not only p-values.
- *Author as content writer:* item quality is checked by an external teacher (Content QA).

10. Project Management and Risks

10.1 Tools and links

Artefact	Location
Git repository	https://github.com/Yerrrzat/Web-platform-UNT-
Submitted version	Git tag `prob-period-1`
YouTrack project	https://yerzat-diploma.youtrack.cloud/projects/UNT (project key **UNT**). The backlog is also exported in `youtrack/tasks.csv`
Documentation	`docs/` (this report), `diagrams/`, `prototype/`

10.2 Roles and responsibilities

This is an **individual project**, so one person has several roles. The supervisor reviews significant changes at agreed checkpoints, as the repository rules require for individual projects.

Role	Person	Responsibilities
Project manager / product owner	Yerzat Qaldyqan	Scope, backlog, priorities, milestones, risk register, stakeholder communication, commercialization analysis
Business analyst	Yerzat Qaldyqan	Interviews, requirements, acceptance criteria
Developer (full-stack)	Yerzat Qaldyqan	Next.js app, database, payments integration, tests, deployment
Content author	Yerzat Qaldyqan	UNT Informatics items, explanations, topic tags
Supervisor / reviewer	Zarina Almabekkyzy	Reviews deliverables at checkpoints, approves scope changes, reviews selected pull requests
External content reviewer	Informatics teacher (to be confirmed)	Checks one version per batch for correctness

10.3 Way of working

- **Process:** Scrum-like 2-week sprints, run solo. Each sprint has a goal, a backlog in YouTrack and a short review note in `docs/sprint-notes/`.
- **Tasks:** every meaningful work item is a YouTrack issue (`UNT-n`) with an assignee, priority, due date, expected outcome and acceptance criteria. A task is set to *Done* only when its acceptance criteria are met.
- **Git workflow:** `main` is the reviewed baseline. Work happens in branches named `UNT-n-short-name`. Commit messages start with the task ID (for example `UNT-8: Add use case diagram`). Changes are merged through a pull request that references the task and describes how the work was checked. Significant PRs (scoring, billing, security) are reviewed by the supervisor at checkpoints.
- **Traceability:** YouTrack task ↔ branch ↔ PR ↔ commit. Each PR link is added back to the YouTrack task.
- **Definition of Done:** acceptance criteria met, tests pass in CI, documentation updated, PR merged into `main`.

10.4 Milestones

Milestone	Target date	Deliverables	Status
M0 Topic approval	Oct 2026	Approved diploma topic	Done

Milestone	Target date	Deliverables	Status
M1 Probationary period 1 (`prob-period-1`)	**24 Oct 2026**	Proposal, evidence, literature review, comparisons, requirements, diagrams, prototype, verification plan, PM plan	In progress (this submission)
M2 Core MVP (internal)	19 Dec 2026	Auth, import, renderer, test session, scoring, results	Planned
M3 MVP ready (supervisor checkpoint)	31 Jan 2027	Billing (test mode), events/dashboard, 10 versions, deployment	Planned
M3a Development complete (supervisor checkpoint)	28 Feb 2027	Testing finished, bugs fixed, release for pilot	Planned
M4 Pre-defense 2	Date set by the school (expected Feb 2027)	Working MVP demo, test reports	Planned
M5 Pilot complete	28 Mar 2027	Pilot data, survey, metrics M1–M9	Planned
M6 Final pre-defense	Date set by the school (expected Apr–May 2027)	Commercialization assessment, final report	Planned

10.5 Backlog snapshot (top items)

ID	Summary	Priority	Due	State
UNT-1 … UNT-11	Analysis, design and planning deliverables (this report)	Critical/Major	23 Oct 2026	Done
UNT-3	Evidence of relevance: 3–5 interviews	Critical	24 Oct 2026	In progress
UNT-12	Supervisor review of probationary-period materials	Critical	24 Oct 2026	To do
UNT-13	Tag `prob-period-1` and submit the form	Critical	24 Oct 2026	To do
UNT-14	Scaffold Next.js app with CI	Critical	7 Nov 2026	To do
UNT-15	Database schema and migrations	Critical	7 Nov 2026	To do
UNT-19 / UNT-20	Test session + scoring engine	Critical	5 Dec 2026	To do
UNT-22	Freemium access and subscription billing	Critical	16 Jan 2027	To do
UNT-27	Pilot with ≥ 30 students	Critical	28 Mar 2027	To do
UNT-28	Commercialization assessment	Critical	15 Apr 2027	To do

The full backlog (28 issues) is in `youtrack/tasks.csv`.

10.6 Risk register

Probability (P) and impact (I) are rated 1–3; score = P × I.

ID	Risk	P	I	Score	Mitigation	Owner
R1	One-person team: workload from other courses delays development	3	3	9	Strict MVP scope (MoSCoW), 2-week sprints, buffer before each milestone, cut *Should/Could* items first	Student
R2	Too few pilot participants (< 30)	2	3	6	Recruit early via teachers and the author's student groups; start recruitment in January; offer free premium access to pilot users	Student

ID	Risk	P	I	Score	Mitigation	Owner
R3	Content errors (wrong keys or explanations) damage trust	2	3	6	Import validator, external teacher review, “report error” button (FR-18)	Student
R4	Unfamiliar stack (Next.js) slows progress	2	2	4	Learning spike in S1–S2; use well-documented libraries; keep a monolith	Student
R5	Payment provider onboarding needs a registered business	3	2	6	Test mode for MVP; `PaymentProvider` abstraction; manual activation fallback; measure intent instead of revenue	Student
R6	Personal data of minors in the pilot	2	3	6	Minimal data, consent forms, pseudonymous analytics, privacy policy, data deletion (NFR-05)	Student
R7	UNT specification changes for 2027	2	2	4	Topic tags stored as data; follow NTC announcements; content mapped to the latest specification	Student
R8	Free NTC trial reduces willingness to pay	2	2	4	Differentiate on explanations and Informatics depth; test pricing in pilot; position as a complement	Student
R9	Hosting free-tier limits or outages during the pilot	1	2	2	Monitor usage; budget up to 10,000 KZT per month; backups of the database	Student
R10	Copyright concerns about test items	1	3	3	Only original items written to the public specification; no NTC items copied	Student

10.7 Dependencies

Dependency	Needed for	Risk link
Supervisor availability for checkpoint reviews	M1, M4, M6	R1
Teachers/schools for interviews and pilot recruitment	UNT-3, UNT-27	R2, R6
Payment aggregator test account	UNT-22	R5
External informatics teacher for content QA	UNT-24	R3
NTC specification for 2027	UNT-24	R7
Hosting/DB providers (Vercel, Neon/Supabase)	UNT-26	R9

10.8 Next steps (after prob-period-1)

1. Fill in the interview table (UNT-3) and record the supervisor’s feedback (UNT-12).
2. Sprint 2 (from 26 Oct): scaffold the app with CI (UNT-14), database schema (UNT-15), hallway test of the prototype (UNT-25).
3. Sprint 3: authentication, import pipeline and item renderer (UNT-16 … UNT-18).

11. Supervisor Feedback

Supervisor: **Zarina Almabekkyzy**

This log records the supervisor’s comments on key deliverables and the decisions taken on them. Each row links to the commit, pull request or YouTrack task with the resulting change.

11.1 Feedback log

#	Date	Deliverable reviewed	Supervisor comment (summary)	Decision	Resulting change (commit / PR / task)	Status
F1	18 Oct 2026	Diploma topic and justification (обоснование)	Agreed to supervise the topic; approval documents prepared for the dean	Accepted	Proposal and scope written for the approved topic (UNT-2, PR #5)	Closed
F2	Oct 2026	Probationary-period package (this report)	Review pending	—	—	Open (UNT-12)

11.2 Agreed checkpoints

Checkpoint	Planned date	What the supervisor reviews
CP1	Before 24 Oct 2026	Probationary-period package (all 11 sections)
CP2	Mid-December 2026	Scoring engine and test-session PRs (UNT-19, UNT-20)
CP3	Late January 2027	Billing/security PR (UNT-22), pilot plan and consent forms
CP4	Before final pre-defense	Pilot results and commercialization assessment